SAN BARTOLOMÉ JHS LA MERCED	TEACHING-LEARNING PLAN	PGF-02-R39 DATE August 12 – November 5, 2026

1. Identification

Area of knowledge: Global Economics
Teacher's name: Nicolas Lopez
Grade: 10
Term: First Term, 2026–2027
Overall implementation period: August 12–November 5, 2026
Institutional code and version: PGF-02-R39, Version 08

2. Learning Objective

LEARNING OBJECTIVE

By the end of the term, students will conduct and communicate an integrated economic decision analysis under uncertainty. Starting from a realistic decision-maker and context, they will distinguish controllable alternatives from the uncertain event and its mutually described states of nature, explain the consequence produced by each alternative–state pair, and represent its direct or calculated payoff. They will calculate quantities, revenue, variable costs, fixed costs and profit; organise complete payoff tables; and justify choices without probabilities using Maximax and Maximin. They will then measure opportunity loss and apply Minimax Regret, use probabilities to calculate Expected Monetary Value (the Bayes criterion), and analyse how a recommendation changes when a probability changes. Finally, they will compare Bayes/expected value, Maximin and Minimax Regret on common evidence, explicitly rank the alternatives, and use those rankings in staged selection rules represented by decision trees. They will then identify a specific realistic application, research information available for that same context, and use the organised real-world data to validate whether the decision-analysis framework can be meaningfully applied.

ASSESSMENT CRITERIA

C1. Identify and interpret the decision-maker, economic context, alternatives, uncertain event, states of nature, consequences and units in a decision under uncertainty.
C2. Organise supplied payoffs and decision elements in a complete payoff table, with alternatives as rows and states of nature as columns.
C3. Calculate quantities, revenue, variable and fixed costs and profit for each alternative–state pair, and construct the corresponding payoff table.
C4. Apply Maximax and Maximin to select and justify optimistic and conservative alternatives without using probabilities.
C5. Construct an opportunity-loss table and apply Minimax Regret to select the alternative with the smallest maximum regret.
C6. Calculate Expected Monetary Value from valid state probabilities and apply the Bayes criterion to select the alternative with the highest expected payoff.
C7. Analyse the sensitivity of an expected-value recommendation to changes in probabilities, including indifference points and preference ranges.
C8. Compare Bayes/expected value, Maximin and Minimax Regret on the same dataset and explain why their recommendations may agree or differ.
C9. Represent and follow staged decisions or criterion-based filtering through a labelled decision tree.
C10. Conduct and communicate an integrated decision analysis that constructs the model, applies relevant criteria, interprets results and supports a justified final recommendation.
Scope note. The current activities establish the C1–C10 sequence above. Some older catch-up files use earlier criterion numbers (Maximin as C5, regret as C6, and expected value as C7). They are used by mathematical skill, not by their legacy number; the current criteria in this plan follow Practice Activities 3–4 and the C8–C10 Lesson 3 material.

3. Conceptual standards of the disciplinary knowledge that supports the narrative's development.

1. **C1—structure of uncertainty.** A decision is made before an uncontrollable event is resolved. Alternatives are available actions; states are possible event outcomes; each ordered pair (A_i, S_j) has a consequence and a payoff with stated meaning and units.
2. **C2—two-dimensional representation.** A payoff table is a complete mapping from alternatives (rows) and states (columns) to $P_{ay}(A_i, S_j)$. A supplied payoff is placed, not recalculated; labels and units preserve the contextual meaning.
3. **C3—economic payoff construction.** Revenue is income and cost is the value of resources used. Variable components equal a per-unit rate times $q(A_i, S_j)$; fixed components are not multiplied by quantity. Profit is revenue minus cost for the same cell.
4. **C4—attitudes without probability.** Maximax compares each row's best payoff and selects the largest; Maximin compares each row's worst payoff and selects the largest. These rules encode optimistic and protective priorities rather than probabilistic averages.
5. **C5—foregone opportunity.** Regret is measured within a state relative to that state's best available payoff. The maximum regret summarizes an alternative's worst foregone opportunity, and Minimax Regret minimizes that summary.
6. **C6—choice under risk.** Probabilities are non-negative and sum to one. $EV(A_i) = \sum_j \Pr(S_j) P_{ay}(A_i, S_j)$ is a probability-weighted monetary average; Bayes selects the greatest EV, not necessarily any single observed payoff.
7. **C7—robustness of assumptions.** In a two-state model, $\Pr(S_1) = p$ and $\Pr(S_2) = 1 - p$. Expected values are linear functions of p; intersections are indifference points and inequalities identify preference ranges on $0 \le p \le 1$.
8. **C8—criterion comparison.** Bayes emphasizes probability-weighted performance, Maximin protects the floor, and Minimax Regret limits foregone opportunity. Agreement strengthens a recommendation; disagreement requires interpretation of priorities and assumptions.
9. **C9—sequential representation.** A decision tree shows an initial candidate set, labelled decision/filter nodes, retained branches and terminal alternatives. The order of staged rules matters and every branch must be traceable to a calculation.
10. **C10—integrated economic argument.** A defensible analysis links context, assumptions, joint or single-state probabilities, payoff construction, criterion results, rankings, staged selection and limitations. The recommendation must follow from reproducible evidence rather than preference alone.

4. Pre-lesson and Context: Pedagogical actions planned by the teacher to motivate students for the term work proposal (presentation of scenarios, narratives, learning objectives and assessment criteria)

Discovering How We Decide
Connection to criterion: Diagnostic preparation for C1; no formal decision rule is named until students have exposed their informal rules.
Pedagogical framework: Discovery learning through notice–classify–represent–justify, followed by an explicit concept synthesis. Universal Design for Learning (DUA) provides multiple means of engagement, representation and expression.
Organization and timing: Groups of three or four; approximately one opening lesson. Launch and individual choice: 8 minutes. Three stations: 12 minutes each. Gallery comparison: 8 minutes. Synthesis and transition: 12 minutes. Timing may be adjusted to the documented school timetable.
Resources: Printed excerpts or teacher-projected versions of Practice Activity 1's Introductory General Problem and Local Café Investment; Practice Activity 2's school-enterprise introductory

problem; sticky notes, calculator, and individual learning notebook. The intake survey may be used only to learn students' preferred working modes, not as criterion evidence.

Expected products: One initial individual recommendation with a reason; one group decision-elements card; one correctly oriented four-cell payoff representation; one revenue–cost explanation; and one revised reflection naming what evidence changed or confirmed the first choice.

Opening vocabulary. *Decision-maker*: person or organisation responsible for the choice. *Alternative*: controllable action selected before uncertainty resolves. *Event*: uncertain phenomenon. *State of nature*: a possible outcome of the event outside the decision-maker's control. *Consequence*: what happens for one alternative–state pair. *Payoff*: numerical value attached to that consequence. *Profit*: revenue minus cost. *Units*: the scale and currency of the payoff.

Opening command words. *Identify* (name from evidence), *distinguish* (show the difference), *organise* (place in a logical structure), *calculate* (show operations), *interpret* (give contextual meaning), and *justify* (support a conclusion with evidence).

Launch—an unrevealed decision rule

Present the Local Café Investment setting from Practice Activity 1 without the worked response: the owner chooses a café format before the market condition is known, and supplied profits differ by alternative and state. Students choose privately and finish: "I chose this because I focused on . . . ". The teacher sorts reasons visibly into "largest possible", "safest low result", "most balanced", and "need more information", without yet naming Maximax, Maximin, regret or EV. This makes prior reasoning observable and creates the need for formal criteria later.

Station 1—Who controls what?

Groups annotate the café scenario and the Practice Activity 2 school-enterprise introductory scenario with six labels: decision-maker, context, alternatives, event, states and consequences. They draw arrows from the choice and the realised state to one consequence. *Expected conclusion:* the decision-maker controls A_i but not S_j; the event (for example, demand) is not itself an alternative; one consequence belongs to each (A_i, S_j). **Support route:** use colour-coded cards headed "chosen before" and "revealed after", sentence frames, and a reduced two-alternative/two-state version. **Extension route:** explain why "profit" is the consequence's measure rather than the uncertain event and identify the units.

Station 2—From narrative to payoff map

Students use Activity 1's Introductory General Problem, with supplied values $P_{ay}(A_1, S_1) = 10$, $P_{ay}(A_1, S_2) = 4$, $P_{ay}(A_2, S_1) = 7$, and $P_{ay}(A_2, S_2) = 8$. On a table-free card they write four lines, one per ordered pair, then arrange alternatives as row headings and states as column headings on sticky notes. *Verified result:* the two rows are $(10, 4)$ and $(7, 8)$. **Support route:** trace the first and second arguments in $P_{ay}(A_i, S_j)$ with different colours and model the first cell. **Extension route:** state what a missing cell would mean and explain why swapping the arguments changes the interpretation.

Station 3—Is the number supplied or built?

Students compare the direct four payoffs above with the Activity 2 introductory school enterprise. For the latter they subtract cost from revenue cell by cell: $(20-9, 13-8)$ for A_1 and $(18-10, 15-9)$ for A_2. *Verified result:* payoff rows $(11, 5)$ and $(8, 6)$, in thousand USD. Groups circle revenue, underline cost, and explain why every subtraction must use information from the same (A_i, S_j). **Support route:** provide the frame $P_{ay}(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$ and a calculator. **Extension route:** decide whether a given lump sum is fixed and why it must not be multiplied by quantity.

Whole-class synthesis and transition

Groups post their products and challenge one classification or cell placement with evidence. The teacher formalises the sequence "choose A_i—observe S_j—experience consequence—measure $P_{ay}(A_i, S_j)$", records the C1 success statement, and contrasts supplied with calculated payoff values. Students revise their launch response: informal preferences are legitimate but cannot be compared consistently until the decision is represented accurately. The next action therefore begins with C1 interpretation, moves to the complete payoff map (C2), builds economic payoffs (C3), and only then applies rules (C4).

5. Experience: Three Pedagogical Actions pedagogical actions that establish the learning objective according to the assessment criteria.

Links to virtual resources or ways used to share experiences (OPEN, TEAMS, etc.) should be included.

PEDAGOGICAL ACTION: N1 **C1–C4** *First Learning Evidence*	**From Decision Contexts to Payoff Tables and Decisions Without Probabilities** **Criteria:** C1–C4 **Cycles:** Institutional cycle allocation not documented: ____________________ **Dates:** Within August 12–November 5, 2026; bundle dates: ____________________ **Instructional hours:** Not documented: ____________________ **Deadline:** ____________________ **Role in progression:** Through mainly collaborative, contextualized real-world decision problems, build a valid model before selecting an alternative. C1 identifies relevant elements and information; C2–C3 determine how that information is organised and used to obtain final payoffs while connecting calculations to economic concepts; C4 uses the resulting payoffs with the first formal rules. This bundle prepares the individual First Learning Evidence. **Materials used:** Practice Activities 1 and 2 (full worked versions, with alternate descriptions for accessible reading); Practice Activity 3's C4 portion; the C1–C4 exam and solution; and aligned C1–C4 catch-up pairs. **Vocabulary.** Decision-maker, context, alternative A_i, event, state S_j, consequence, direct payoff, quantity $q(A_i, S_j)$, revenue, variable rate, fixed component, cost, profit, payoff table, Maximax, Maximin, optimistic, conservative, tie and limitation. **Command words.** Identify, characterise, distinguish, organise, construct, calculate, apply, select, interpret and justify.

Part 1—C1: Interpret the Anatomy of a Decision

Objective: Given an economic narrative, identify every decision element and explain the timing and economic meaning of one consequence.

Classroom format: Teacher think-aloud on Activity 1's Introductory General Problem; groups annotate Local Café Investment and Green Energy Production Mix and transfer the method to Activity 2's school-enterprise or café context; a short individual check confirms each student's understanding.

Theory and procedure. A choice is under uncertainty when an A_i is selected before the event reveals S_j. Students use a fixed audit: (1) name the decision-maker and objective; (2) list only controllable alternatives; (3) name the single uncertain event; (4) list its states; (5) describe what the pair (A_i, S_j) causes; (6) state payoff meaning, sign and units. The event is the variable phenomenon; its states are the outcomes. A consequence is richer than its numerical payoff.

Common errors. Treating high demand as an alternative; listing the payoff as a state; failing to name who decides; mixing actions and outcomes; describing the consequence without units; or assuming the later state can be chosen. **Practice and exact sources.** Activity 1, Introductory and Advanced General Problems and Direct Payoff Models (Local Café, Green Energy, Global Shipping); Activity 2, Introductory General Problem. Students complete decision-element profiles, not merely copied lists. The exam's Weekend Market Stall and Urban Delivery Service provide an ungraded collaborative rehearsal. The solution verifies that, for the Weekend Market, the cooperative chooses the location, demand is strong or weak, and each cell is weekend profit.

DUA support. Offer alternate-description passages, a before/after timeline, icons, oral explanation or labelled sketch, and sentence starters. **Extension.** Analyse an Activity 1 complete four-component context and explain which data vary by alternative, state, or both. **Evidence.** One annotated narrative, one six-element profile and one paragraph interpreting a specific $P_{ay}(A_i, S_j)$.

Part 2—C2: Organise Supplied Payoffs

Objective: Construct a complete and correctly oriented payoff table from supplied direct or component-based payoffs.

Classroom format: Group card sort and construction using realistic contexts, peer audit between groups, then a short individual check.

Theory and procedure. In current notation, $P_{ay}(A_i, S_j)$ is the payoff when row alternative A_i is selected and column state S_j occurs. Students (1) establish row and column labels; (2) retain the stated units; (3) match the first function argument to the row and the second to the column; (4) place every supplied value; (5) count $m \times n$ cells; and (6) interpret one cell in context. When components rather than final values are supplied, Activity 1 uses

$$P_{ay}(A_i, S_j) = q(A_i, S_j)[v(A_i) + v(S_j)] + f(A_i) + f(S_j),$$

with absent components treated as zero.

Common errors. Transposing alternatives and states, reversing the arguments, adding probabilities to a method that does not need them, omitting a cell, multiplying a fixed component by quantity, or silently changing units. **Practice and exact sources.** Activity 1's Introductory General Problem (verified rows $(10, 4)$ and $(7, 8)$), Advanced General Problem (verified rows $(12, 19)$ and $(23, 33)$), and the Direct through Complete model ladder. Use Local Café and one teacher-selected variable/fixed pair; use Weekend Market Stall as the final individual rehearsal. **DUA support.** Colour the two function arguments, provide a blank cell checklist, reduce the first case to 2×2, and allow manipulable row/column cards. **Extension.** Diagnose and correct a transposed table and justify whether two tables encode the same mapping. **Evidence.** Two complete payoff representations, visible cell work and two contextual cell interpretations.

Part 3—C3: Calculate Revenue, Cost and Profit

Objective: Build every payoff from direct totals or from variable and fixed revenue/cost components before organising the results.
Classroom format: Teacher model, calculation teams assigned different cells, jigsaw verification, and a short independent construction after the collaborative contextualized work.
Theory and procedure. Use $q(A_i, S_j)$ for cell quantity. Activity 2 models revenue and cost independently:

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)[r^A(A_i) + r^S(S_j)] + F^{R,A}(A_i) + F^{R,S}(S_j), \\ C(A_i, S_j) &= q(A_i, S_j)[c^A(A_i) + c^S(S_j)] + F^{C,A}(A_i) + F^{C,S}(S_j), \\ P_{ay}(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

For each cell students separate revenue from cost, classify variable/fixed and alternative/state components, substitute quantity with units, total revenue, total cost, subtract, and check sign and scale. Direct total revenue or cost is not decomposed again.
Common errors. Adding cost to revenue, subtracting only variable cost, multiplying fixed terms by q, using one state's quantity in another cell, combining dollars with thousand dollars, or rounding before the final result. **Practice and exact sources.** Activity 2's Introductory General Problem has verified payoff rows $(11, 5)$ and $(8, 6)$. Progress from its direct model to one- and two-component contexts, then to the Complete Four-Component Model. Use School Lunch Service as a short rehearsal: verified profits are $(14, 6)$ and $(9, 7)$ thousand USD. Use Cold-Chain Distribution to practise the complete method; its exam solution models all revenue and cost contributions before subtraction. **DUA support.** A revenue/cost sorting mat, one worked cell, calculator, units checklist and partially completed substitutions. **Extension.** Audit a peer's four-component model and locate the first incorrect term rather than only correcting the final payoff. **Evidence.** A full cell ledger, complete payoff representation and written unit check.

Part 4—C4: Decide Without Probabilities

Objective: Apply both Maximax and Maximin and interpret their different attitudes toward uncertainty.
Classroom format: Human row scan, group calculation and structured debate about the realistic decision, followed by an individual exam-style response.
Theory and procedure. For every alternative calculate its row maximum and row minimum:

$$A_{\text{Maximax}} = \arg\max_i \max_j P_{ay}(A_i, S_j), \qquad A_{\text{Maximin}} = \arg\max_i \min_j P_{ay}(A_i, S_j).$$

Students mark one best and one worst value per row, compare like summaries, report ties, name the selected alternative, and interpret the upside or protection. Displayed probabilities are ignored for these two rules.
Common errors. Taking a column maximum, selecting the smallest row minimum, averaging, using probabilities, reporting only a number, or describing Maximin as minimizing payoff. **Practice and exact sources.** Activity 3, General 2×2 and contextual problems of increasing dimensions; exam Mobile Produce Shop and Warehouse Technology. Verified exam logic: Mobile Produce gives Maximax Business District (48) and Maximin Residential District (24); Warehouse Technology gives Maximax Automated (96) and Maximin Flexible (50). **DUA support.** Mark row maxima and minima with distinct symbols, verbalise "best of the best" and "best of the worst", and provide a two-stage checklist. **Extension.** Find a case in Activity 3 where the rules disagree and explain the decision-maker attitude each recommendation serves. **Evidence.** Marked row summaries, both selections and a two-sentence comparison.

Practice and Assessment—First Learning Evidence

Practice proceeds from worked examples to paired, then independent problems. Before grading, groups complete an ungraded mixed rehearsal using Weekend Market, School Lunch, Mobile Produce and one complete-component calculation; the teacher returns criterion-coded feedback (C1 meaning, C2 placement, C3 model/arithmetic, C4 rule/interpretation). The **First Learning Evidence** is the repository **individual written exam assessing C1–C4**: decision characterisation, supplied-payoff organisation, direct and multi-component profit calculations, and Maximax/Maximin selections. Students submit the exam with all working. The teacher uses its solution to verify methods and results, then runs correction stations by criterion. Students needing recovery complete the matching C1–C4 catch-up skill set and submit corrected reasoning; solution files are teacher keys, not answer-copy sheets. The resulting payoff fluency is the prerequisite for regret and probability in Action N2.

Exit Ticket N1

- ☐ I can identify the decision-maker, alternatives, event, states, consequences and units.
- ☐ I can place every $P_{ay}(A_i, S_j)$ in the correct payoff position.
- ☐ I can separate revenue and costs, classify variable and fixed components, and calculate profit.
- ☐ I can apply and interpret Maximax and Maximin without probabilities.
- ☐ I can show full reasoning and I am ready for the C1–C4 written exam.

Independent study: Complete assigned Activity 1–3 problems in the shared course channel or learning notebook as directed. If OPEN, Teams or OneNote is used by the class, upload a readable page showing labels, substitutions, units and conclusions; an answer alone is not evidence.

Required student evidence: Discovery products; C1 profile; C2 payoff constructions; C3 calculation ledger; C4 row summaries and interpretations; ungraded rehearsal; exit ticket; written exam; and, when assigned, criterion-specific corrected catch-up work.

PEDAGOGICAL ACTION: N2 **C5–C7** *Second Learning Evidence*	**Regret, Expected Value and Probability Sensitivity** **Criteria:** C5–C7 **Cycles:** Institutional cycle allocation not documented: ____________________ **Dates:** Within August 12–November 5, 2026; bundle dates: ____________________ **Instructional hours:** Not documented: ______________ **Deadline:** ____________________________ **Role in progression:** Continue collaborative work on contextualized real-world problems while extending the toolkit in order: C5 Minimax Regret, C6 Expected Value/Bayes with probabilities, and C7 sensitivity to changes in probabilities or parameters. Student-created interactive web tools make changing inputs, calculations, rankings and recommendations visible without replacing mathematical reasoning. **Materials used:** Practice Activity 3's C5 sections; Practice Activity 4; the archived C6 supplementary activity and solution and Lesson 2 worked material; the legacy regret (C6) and expected-value (C7) catch-up pairs, remapped by skill; Activity 4 itself for sensitivity remediation. No C5–C7 formal exam source is currently stored. **Vocabulary.** State best, opportunity loss, regret, maximum regret, Minimax Regret, probability, risk, mutually exclusive states, Expected Monetary Value, Bayes criterion, subjective probability, sensitivity, parameter p, indifference point, preference interval and dominated/never-optimal alternative. **Command words.** Construct, subtract, determine, minimise, calculate, weight, compare, solve, analyse, test, interpret and justify. **Interactive web-tool strand.** In groups, students create and use interactive web tools that automatically solve the decision problems studied in this action while allowing probabilities, payoffs or other parameters to be modified. They first show and verify the mathematical work themselves, then use the tool to explore how input changes affect regret or EV calculations, rankings and final decisions. The tool is a manipulable model for conceptual understanding and checking; it does not replace students' formulas, reasoning or contextual interpretation.

Part 1—C5: Opportunity Loss and Minimax Regret

Objective: Construct a regret representation from a payoff table and select the smallest maximum regret.

Classroom format: State-column teams, paired audit, whole-class comparison with C4, and group use of the interactive tool before an independent transfer.

Theory and procedure. For each state first find $b_j = \max_i P_{ay}(A_i, S_j)$. Then compute

$$OL(A_i, S_j) = b_j - P_{ay}(A_i, S_j), \qquad A_{\text{MMR}} = \arg\min_i \max_j OL(A_i, S_j).$$

Thus the best payoff within each state has zero regret. Students find column bests, subtract every payoff from the corresponding state best, find each row's maximum regret, and select the smallest of those maxima.

Common errors. Using a global rather than state-specific best, reversing the subtraction and creating negative regret, applying probabilities, selecting the largest maximum regret, or confusing loss payoff with opportunity loss. **Practice and exact sources.** Activity 3 General 2×2 and the Community Theater, Holiday Inventory and Routing System problem types; then the supplementary Cafeteria Supply or Bookstore Inventory case. The supplementary solution verifies state maxima and each subtraction. Activity 3 examples also show that Minimax Regret may select a different alternative from Maximax/Maximin. **DUA support.** Highlight one state at a time, use the sentence "best in this state minus this payoff", and provide a zero-regret check. **Extension.** Explain why adding the same constant to every payoff within a state leaves regrets unchanged. **Evidence.** State-best list, complete regret work, row maximum regrets and contextual choice.

Part 2—C6: Expected Monetary Value and Bayes Choice

Objective: Validate probabilities, calculate EV for every alternative, rank alternatives and interpret the Bayes choice.

Classroom format: Group probability check, teacher model, paired parallel calculations, and tool-assisted parameter exploration before an individual comparison.

Theory and procedure. Confirm $0 \le p_j \le 1$ and $\sum_j p_j = 1$. Then

$$EV(A_i) = \sum_j \Pr(S_j) P_{ay}(A_i, S_j),$$

using the same probability for every alternative in a state. Show each product, sum without premature rounding, compare all EVs, report ties, and interpret EV in the stated monetary units. EV is a long-run or probability-weighted measure, not a guaranteed payoff.

Common errors. Probabilities not summing to one; matching a probability to an alternative rather than a state; choosing the largest single payoff; omitting negative payoffs; dividing the sum again; or calling EV guaranteed. **Practice and exact sources.** Activity 4 C6: Community Theater (verified $EV_A = 40$, $EV_B = 38.4$, $EV_C = 37$, select A); Holiday Inventory (51, 54.5, 48.5, select B); Routing System (20.75

versus 22.4, select B), all in thousand dollars. The legacy C7 catch-up and Lesson 2 material provide extra expected-value cases by skill. **DUA support.** Probability strip, product-by-product template, calculator and verbal interpretation frame. **Extension.** Compare the Bayes choice with C4/C5 choices on the same Activity 3 dataset and identify which added information changed the analysis. **Evidence.** Probability audit, complete EV expressions, ranking and recommendation with a non-guarantee statement.

Part 3—C7: Probability Sensitivity

Objective: Express EV as a function of p, find valid tie points and identify the preferred alternative on each probability interval.
Classroom format: Graph-free algebra model, paired inequality check, interactive-tool exploration of changing probabilities or parameters, number-line gallery, and independent explanation.
Theory and procedure. For two states set $\Pr(S_1) = p$ and $\Pr(S_2) = 1 - p$, where $0 \le p \le 1$. Students form and simplify every EV function, equate competing functions to locate an indifference point, test one value on each side, state exact preference ranges, and check whether another alternative is ever the upper EV choice. Endpoint checks detect reversed inequalities.
Common errors. Assigning p to both states; forgetting $1 - p$; solving a tie but not preference ranges; accepting p outside $[0, 1]$; reversing above/below; or assuming every plan must be optimal somewhere.
Practice and exact sources. Activity 4 C7 Marketing Options: $EV(A) = 12p + 12$, $EV(B) = 2p + 16$, tie $p = 0.4$, A above and B below. Production: $EV(A) = 28p + 4$, $EV(B) = 6p + 14$, $EV(C) = 2p + 10$; A/B tie at $5/11$, A above, B below, and C never optimal. Zoning: $EV(A) = 20p + 6$, $EV(B) = 8p + 12$, tie 0.5, A above and B below. **DUA support.** Colour p and $1 - p$, use a three-line tie/test/conclude organizer and allow number-line or oral output. **Extension.** Prove C is never optimal in Production by comparison and explain the economic meaning of a probability estimate close to a threshold. **Evidence.** Simplified EV functions, algebraic tie, interval statement, test values and interpretation of robustness.

Practice and Assessment—Second Learning Evidence

Students first complete ungraded collaborative regret, EV and sensitivity rounds using the named contexts. Peers audit one operation at a time; teacher feedback is coded C5 (benchmark/subtraction/selection), C6 (probability/product/sum/interpretation), or C7 (function/tie/interval). The **Second Learning Evidence** is an **individual written exam assessing C5–C7** based on the C5–C7 practice activities, with a supplied payoff table for regret, a probability table for EV, and a two-state parameter problem for sensitivity. Its repository file, date, duration, point allocation and threshold are not documented and remain institutional completion items; none is invented here. Students submit the exam and all calculations. Corrections are made by criterion. For regret, the archived supplementary and legacy C6 catch-up are used; for EV, the legacy C7 catch-up is used; for sensitivity, students redo the relevant Activity 4 case with changed prompts and explain the threshold. Corrected evidence is attached to the original. The bundle supplies the three rules compared in N3.

Exit Ticket N2

- ☐ I can find each state's best payoff and calculate non-negative opportunity losses.
- ☐ I can select the smallest maximum regret and justify the choice.
- ☐ I can verify probabilities, calculate every EV and apply Bayes.
- ☐ I can find a probability tie point and state preference ranges on $0 \le p \le 1$.
- ☐ I can show full reasoning and I am ready for the C5–C7 written exam.

Independent study: Finish the assigned Activity 3 regret and Activity 4 EV/sensitivity cases. Submit through the class's established OPEN/Teams/OneNote route if directed, or in the learning notebook. Evidence must include state maxima, products, sums, algebra, valid probability ranges and written conclusions.
Required student evidence: C5 regret analysis; three C6 EV calculations; one fully explained C7 threshold analysis; a student-created interactive web tool and documented parameter explorations; collaborative audit; exit ticket; individual written exam; criterion-coded correction; and assigned remediation.

PEDAGOGICAL ACTION: N3 **C8–C10** *Third Learning Evidence*	**Ranking Alternatives, Decision Trees and Real-World Application** **Criteria:** C8–C10 **Cycles:** Institutional cycle allocation not documented: ____ **Dates:** Within August 12–November 5, 2026; bundle dates: ______________________________________ **Instructional hours:** Not documented: _____________________________ **Deadline:** ____ **Role in progression:** Begin with structured recall of previously studied methods; use C8 to compare methods and explicitly rank alternatives; use those rankings in C9 decision trees and staged processes; then use C10 to present and analyse a real-world context in which the term's techniques can be meaningfully applied. The final evidence has a C8–C9 individual exam followed by C10 validation research on exactly the context each student selects in the exam's final question. **Materials used:** Archived Lesson 3 Material, the only current C8–C10 worked source, plus selected Activity 3–4 datasets for structured recall and retrieval. No separate current C8–C9 exam or C10 validation-research brief is physically present. **Vocabulary.** Common dataset, ranking, criterion agreement, criterion conflict, decision rule, candidate set, filter, node, branch, leaf, staged decision, joint state, joint probability, independence assumption, integrated analysis, evidence, limitation and recommendation. **Command words.** Apply, rank, compare, contrast, represent, trace, integrate, evaluate, communicate, defend and recommend.

Part 1—Structured Recall and C8: Compare Methods and Rank Alternatives

Objective: Recall the studied methods, apply Bayes, Maximin and Minimax Regret to one dataset, provide ordered results, and explain agreement or conflict in economic terms.

Classroom format: Structured retrieval of C4–C7 methods and ranking directions, then three expert groups (one criterion each), mixed-group reconciliation, and individual synthesis.

Theory and procedure. A fair comparison holds the alternatives, states, payoffs and, for Bayes, probabilities constant. Students first recall when and how each method is used. They then validate the data, calculate EVs and rank greatest to least; calculate row minima and rank greatest to least; construct regrets and rank maximum regrets least to greatest; then state each winner and explain what each rule values. A difference is not a calculation failure if the verified rankings differ.

Common errors. Comparing results from different datasets, ranking regret in the wrong direction, using probability in Maximin/regret, reporting only winners rather than complete rankings, or claiming one rule is universally correct. **Practice and exact sources.** Lesson 3 Material C8 Problems 1 and 2. In its first worked dataset, Bayes and Maximin select A_2, while Minimax Regret ties A_1 and A_3; students reproduce the calculations and interpret why rules differ. Retrieve one Activity 4 dataset and add the non-probability rules for transfer. **DUA support.** Criterion cards with direction arrows, role-based group calculation, ranking sentence frames and oral recorded comparison. **Extension.** Identify what change in payoff or probability could alter only Bayes and justify the claim. **Evidence.** A structured-recall record, three auditable rankings, winner comparison and an assumption-aware paragraph.

Part 2—C9: Use Rankings in a Staged Decision Tree

Objective: Build and follow a labelled tree that uses C8 rankings to filter a candidate set through a specified sequence of criteria.

Classroom format: Teacher modelling, floor-sized branch sequence, paired redraw, individual trace.

Theory and procedure. In the repository's Lesson 3 convention, the tree is a staged filter rather than a chance-node EV tree. Start at D_0 with all alternatives; label $D_1, D_2, \ldots$ with the criterion, its ranking and retention rule; branch only to retained candidates; show terminal leaves and identify the final leaf. Calculations and explicit rankings must precede the diagram. Because filters reduce a set, the specified order can affect the final choice.

Common errors. Drawing a decorative tree without values or rankings, removing an alternative not excluded by the rule, applying a later criterion to the full set, changing the given filter order, missing ties, or confusing a leaf with a state of nature. **Practice and exact sources.** Lesson 3 C9 problems: one process keeps the best four EV options, then the best two Maximin options, then the smallest Minimax Regret; its verified final selection is B_2. Another keeps three by EV, two by Minimax Regret and selects by Maximin; its verified final selection is X_5. Students reproduce one and design a clear text-tree for the other. **DUA support.** Movable candidate cards, node template, colour-coded retained branches, and permission to submit a labelled linear text tree. **Extension.** Swap two stages, recalculate the candidate sets and determine whether the outcome is order-sensitive. **Evidence.** Candidate-set record, criterion rankings, labelled nodes/branches/leaves and a trace explaining the selected terminal alternative.

Versión 08

Part 3—C10: Validate a Student-Selected Real-World Application

Objective: Research the specific real-world situation proposed in the final C8–C9 exam question, organise realistically obtainable data, and use it to validate whether the studied decision-analysis framework is meaningfully applicable.

Classroom format: Individual research on each student's exam-selected context, with group research workshops, source/data-organisation peer review and teacher conferences.

Theory and procedure. No unrelated teacher-assigned topic is introduced. Each student continues exclusively with the situation identified in the final exam question and follows a validation protocol:

1. retain the exam-selected decision-maker and situation, refining its objective, alternatives, event/states, consequences, units and assumptions where the research supports doing so;
2. obtain and cite relevant real-world information or data that could realistically inform payoffs, probabilities or parameters, and organise it transparently;
3. distinguish observed data from estimates and document gaps, limitations and any assumptions, including joint probabilities only when the evidence supports multiplication;
4. use the organised information to test how the studied payoff, ranking and staged-decision methods fit the situation; and
5. conclude whether and within what limits the decision-analysis framework is meaningfully applicable to this context.

Joint states in the Lesson 3 food-processing model combine demand and operating cost probabilities under the worked independence structure; students must not assume independence in their selected case unless research evidence supports it or the assumption is explicitly justified. These examples model technical analysis but do not supply a substitute topic for the student's research.

Common errors. Changing to an unrelated topic after the exam, collecting information that does not inform the selected decision, treating estimates as observed facts, multiplying unrelated probabilities without an assumption, mixing revenue and profit, hiding data organisation, or claiming applicability without identifying limitations. **Practice and exact sources.** Lesson 3 C10 Food-Processing Packaging provides a worked application: verified joint probabilities are $0.30, 0.18, 0.12, 0.20, 0.12, 0.08$; verified EV ranking is $P(220) > R(213) > Q(205)$, Maximin is $P(60) > R(35) > Q(5)$, maximum regrets are $P = 10, R = 25, Q = 55$, and the staged tree selects P. The Online Retailer extension provides a six-option analysis; its verified final leaf is S_5. **DUA support.** Research and data-organisation checklist, modelled source-to-variable example, calculator/spreadsheet as a checking tool, oral or written planning options, and frequent teacher conferences. **Extension.** Test a changed probability or parameter supported by the research and explain whether the applicability conclusion remains robust. **Evidence.** The final exam response naming one specific real-world situation and realistically obtainable data; cited research confined to that situation; an organised data record; an applicability analysis linking the information to the studied framework; and an individual conclusion stating evidence, assumptions and limitations.

Practice and Assessment—Third Learning Evidence

An ungraded collaborative Lesson 3 rehearsal precedes formal evidence: expert groups verify one method, exchange results, reconcile rankings on the same dataset and trace those rankings through a tree. Teacher feedback codes C8 (completeness/direction/interpretation) and C9 (node/filter/trace). The **Third Learning Evidence** has two connected phases. **Phase 1 is an individual written exam focused specifically on C8 and C9**: students compare methods, explicitly rank alternatives, and apply the results through decision trees. Its final question is a thinking-and-writing task in which each student proposes one specific real-world situation where the methods studied in class could be applied and identifies relevant real-world data that could realistically be obtained for it. **Phase 2 is C10 validation research exclusively on the topic that the student selected in that final exam question**: the student obtains and organises the relevant real-world information/data and uses it to validate the applicability and limits of the decision-analysis framework. There is no separate or unrelated teacher-assigned project topic. Exam and validation-research files, dates, duration, point allocation and thresholds are not present in the current directory and remain blank institutional items. Feedback for Phase 1 is returned by C8–C9; feedback for Phase 2 addresses the relevance and organisation of evidence and the C10 applicability analysis.

Exit Ticket N3

- ☐ I can recall the studied methods and rank one dataset using Bayes, Maximin and Minimax Regret.
- ☐ I can explain agreement or conflict among the rankings.
- ☐ I can use those rankings to build and trace a staged decision tree with labelled filters and leaves.
- ☐ I can propose a specific real-world application and identify relevant data that could realistically be obtained for it.
- ☐ I can organise researched data from that same context and use it to analyse the framework's applicability and limitations.
- ☐ I am ready for the individual C8–C9 exam and the connected C10 validation phase on my exam-selected context.

Independent study: Recalculate one Lesson 3 example without viewing its worked section, then self-audit. After the exam, research only the context selected in its final question and organise relevant obtainable information/data with sources, assumptions and limitations. Digital submission occurs only through the class's established OPEN, Teams or OneNote route; every result must be traceable to formulas, rankings, researched information and written interpretation.

Required student evidence: Structured recall; C8 comparison and rankings; C9 tree and trace; ungraded rehearsal; exit ticket; individual C8–C9 written exam including the final real-world-context response; and C10 research, organised data and applicability validation exclusively for that exam-selected context.

6. Learning-Evidence Overview and Institutional Completion

First Learning Evidence Individual written C1–C4 exam stored in the repository. Practice: Activities 1–3. Submission: exam with reasoning. Feedback and recovery: criterion-coded correction and aligned C1–C4 catch-up. Connection: produces reliable payoff tables for N2.

Second Learning Evidence Individual written C5–C7 exam based on Activities 3–4. Submission: regret, EV and sensitivity work with full reasoning. Feedback and recovery: criterion-coded review and skill-aligned supplementary/catch-up practice. Connection: supplies all rules used comparatively in N3. *Exam file and administrative details are not documented.*

Third Learning Evidence Two connected phases: (1) an individual C8–C9 written exam comparing and ranking alternatives and applying the results through decision trees, ending with the student's proposal of one specific real-world application and identification of realistically obtainable relevant data; and (2) C10 research exclusively on that exam-selected topic, organising the real-world information/data and using it to validate the framework's applicability. Submission: exam workings and the connected C10 research/data/applicability analysis. Feedback and recovery: C8–C9 criterion-coded re-analysis with Lesson 3 material and C10 feedback on evidence relevance, organisation and validation. *Current exam/validation-phase files and administrative details are not documented.*

Unresolved institutional fields. Exact cycle numbers, bundle dates, instructional hours, deadlines, grading thresholds, and the administrative specifications for Learning Evidences 2 and 3 must be completed by the school. The stored C1–C4 exam states a threshold internally, but this plan does not generalise that file-specific statement into an undocumented term-wide rule.

References:

All paths are relative to `book2026-2027/grade10/term_1/`. Only inspected materials actually used in this plan are listed. Archived files are identified as such and are used for worked extension or skill-aligned support, not treated as current formal assessments.

- **Course map:** `README.md`.
- **Activity 1, full and accessible parallel text:** `1-practice_activities/G10_T1_L1_C1C2_practice_activity1.tex`; `1-practice_activities/G10_T1_L1_C1C2_practice_activity1-alternate-descriptions.tex`.
- **Activity 2, full and accessible parallel text:** `1-practice_activities/G10_T1_L1_C1C3_practice_activity2.tex`; `1-practice_activities/G10_T1_L1_C1C3_practice_activity2-alternate-descriptions.tex`.
- **Activities 3–4 and teacher map:** `1-practice_activities/G10_T1_L2_C4C5_practice_activity3.tex`; `1-practice_activities/G10_T1_L2_C6C7_practice_activity4.tex`; `1-practice_activities/G10_T1_practice_activities_summary.tex`.
- **First written evidence and verified key:** `2-learning_evidences/G10_T1_L1_C1C2C3C4_exam.tex`; `2-learning_evidences/G10_T1_L1_C1C2C3C4_exam_solution.tex`.
- **C1–C4 recovery pairs:** `3-catch_ups/G10_T1_C1_catchUp.tex`; `3-catch_ups/G10_T1_C1_catchUp_soution.tex`; `3-catch_ups/G10_T1_C2_catchUp.tex`; `3-catch_ups/G10_T1_C2_catchUp_soution.tex`; `3-catch_ups/G10_T1_C3_catchUp.tex`; `3-catch_ups/G10_T1_C3_catchUp_soution.tex`; `3-catch_ups/G10_T1_C4_catchUp.tex`; `3-catch_ups/G10_T1_C4_catchUp_soution.tex`.
- **Skill-aligned older recovery pairs:** `3-catch_ups/G10_T1_C6_catchUp.tex`; `3-catch_ups/G10_T1_C6_catchUp_soution.tex` (regret); `3-catch_ups/G10_T1_C7_catchUp.tex`; `3-catch_ups/G10_T1_C7_catchUp_solution.tex` (expected value).
- **Archived regret supplement and key:** `4-extra_material/old/G10_T1_L2_C6_activity_supplementary.tex`; `4-extra_material/old/G10_T1_L2_C6_activity_supplementary_solution.tex`.
- **Archived worked Lesson 2 reference:** `4-extra_material/old/G10_T1_L2_MATERIAL.tex`.
- **Archived C8–C10 comparison, tree and integrated-analysis source:** `4-extra_material/old/G10_T1_L3_MATERIAL.tex`.
- **Opening learner-profile support:** `4-extra_material/G10_T1_survey.tex`.